

Python: Weather Forecasting Service

Implement a weather forecasting service in Python with custom exceptions for error handling. The implementation should demonstrate proper exception handling techniques and provide detailed feedback to users when errors occur during weather data processing.

Class Functionality: * **WeatherService** processes weather data for cities, handling errors related to temperature and forecast data by raising specific exceptions as needed. It supports querying weather data for a city based on specified parameters.

1. **weather_data_enum(data):**

Parameters:

- data (dictionary): Data used to create an enumeration.

Return Value: Enumeration object created from the provided data.

process_weather_data(WeatherData, city, query):

Parameters:

- WeatherData (enum): Enum containing weather data for various cities.
- city (string): The city whose weather data is queried.
- query (string): Specific parameter queried (e.g., temperature or forecast).

6. **Return Value:** Returns a string with the queried weather data: "The {query} for {city} is {weather_value}".

Exceptions Raised with Messages: * **InvalidTemperatureData:** Raised if temperature data is missing or invalid. Message: "Invalid or missing {query} data". * **InvalidForecastData:** Raised if forecast data is missing or invalid. Message: "Invalid or missing {query} data". * **InvalidCityName:** Raised if the city name is not in the data. Message: "Invalid city name {city}".

TemperatureOutOfRange: Raised if temperature is outside the valid range (-20 to 50 inclusive). Message: "Temperature {weather_value} is out of range".

Constraints * There are 50 or fewer cities with their weather data.

Input Format For Custom Testing Input from stdin will be processed as follows and passed to the function:

The first line is the number of weather data points that are used for service. The weather data is given afterward, following the format {city}, {temperature}, {forecast}. The next line is the number of queries. Query information follows in the format {city}, {weather_type(temperature or forecast)}.

Sample Case 0 Sample Input For Custom Testing

```
7
New York,25,sunny
London,18,cloudy
Paris,,sunny
Madrid,29,clear
Berlin,,snowfall
Ulaanbaatar,-30,cold
Sydney,30,sunny
2
New York,temperature
Paris,temperature
```

Sample Output

```
The temperature for New York is 25
Error: Invalid or missing temperature data
```

Explanation There are 7 weather data points given, then 2 queries. The result for the first query displays the temperature in New York, but Paris is missing temperature data.

Sample Case 1 Sample Input For Custom Testing

```
7
New York,55,sunny
London,,sunny
Paris,27,
Madrid,,humid
Berlin,14,
Ulaanbaatar,-30,
Sydney,30,sunny
3
New York,temperature
Paris,forecast
xyz,temperature
```

Sample Output

```
Error: Temperature 55 is out of range
Error: Invalid or missing forecast data
```

Error: Invalid city name xyz

Explanation Each query raises the appropriate error.